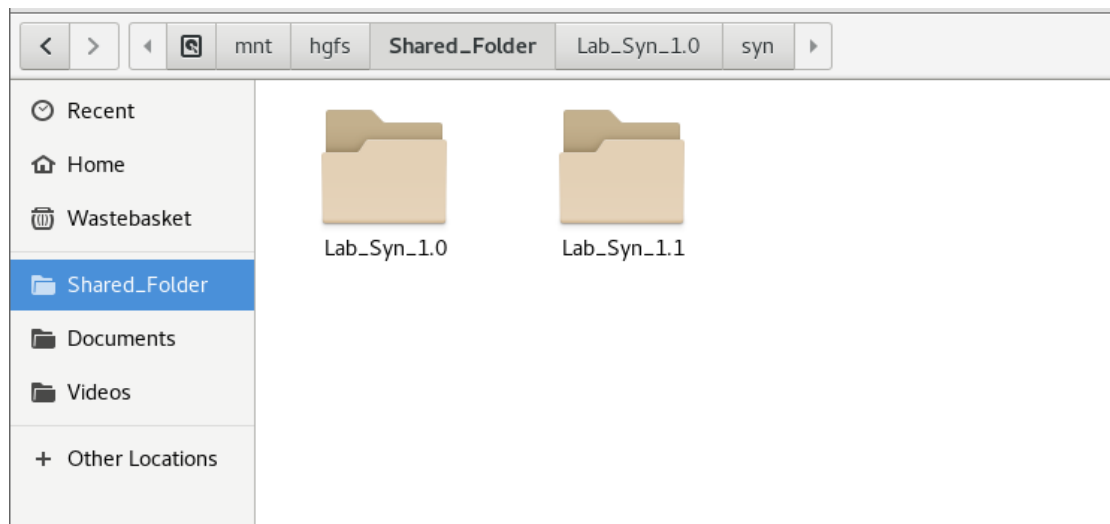


Lab_Syn_1.1

You have to synthesize **Up_Dn_Counter.v** file using **Design Compiler** shell and generate **Technology Dependent** gate level netlists in Verilog format using standard cell library (**scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db**)

Steps: -

1. Place **Lab_Syn_1.1** folder into your shared folder (**Shared_Folder**) in your computer. You will find **Lab_Syn_1.1** appeared inside your virtual machine shared folder in this path (**/mnt/hgfs/Shared_Folder**).



2. Create a script with name **syn_script.tcl** and write the following commands
 - 1) Add the std cell libraries path to the search path
 - 2) Add the rtl files path to the search path
 - 3) Define the values of **target_library** and **link_library** variables to load standard cell libraries (typical)
 - 4) Read the **Up_Dn_Counter.v** file using **read_file** command
 - 5) Link the design & tech files using **link** command
 - 6) Map & Optimize the design using **compile** command
 - 7) Generate the mapped gate level netlist using **write_file** command

```
##### Design Compiler Library Files #####

lappend search_path /mnt/hgfs/Shared_Folder/Lab_Syn_1.1/std_cells
lappend search_path /mnt/hgfs/Shared_Folder/Lab_Syn_1.1/rtl

set TTLIB "scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db"

## Standard Cell libraries
set target_library [list $TTLIB]

## Standard Cell & Hard Macros libraries
set link_library [list * $TTLIB]

##### Reading RTL Files #####

read_file -format verilog Up_Dn_Counter.v

link

##### Mapping and optimization #####

compile

#####
# Write out Design after initial compile
#####

write_file -format verilog -output Up_Dn_Counter_netlist.v

#start_gui

#exit
```

3. Open the terminal inside path (/mnt/hgfs/Shared_Folder/Lab_Syn_1.1/syn)
4. Run the script through design compiler shell using the following command in the terminal and press Enter: -

- dc_shell -f syn_script.tcl
- dc_shell -f syn_script.tcl > syn.log &

